

Linear Algebra (EC) Syllabus

Spring semester 2026

Course Title	Linear Algebra (EC)	Course Code		MC3111		Credit	3	Classroom	형설관 407
		Type		Advanced Required		Hours	4		
Teacher	Name	Class no.	Office	Office hours*	Contact		E-mail		
	임종렬	EC1	형설관 508	Tue3, Wed3, Fri3	051-606-2204		limjongryul@ksa.hs.kr		

*Office Hours: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

1. Course Description

Learning linear algebra is closely related to the study of vectors and linear transformations whereas most of students think that it is about matrices and systems of linear equations. This course intends to provide a thorough understanding of many important key concepts in linear algebra that would be expected to be in most first or second year university courses in the subject. After taking this course, the students should be able not only to explain concepts in linear algebra, but also to carry out computations using the methods they learned from the course.

2. Core Target Competencies

Subject Core Competencies					
Problem solving	Reasoning	Communication	Attitudes and practices	Problem discovery and definition	Information ethics
Solving problems related to fundamental topics of mathematics, Proving and disproving statements; Using correct rules of logic, Write mathematical reasonings in a clear and rigorous way, Learning course material in an organized way, Modelling of concrete problems using mathematics, Use critical thinking to analyze claims.					

3. Textbooks and References

A. Textbook: Linear Algebra: Concepts and Methods by M. Anthony and M. Harvey

B. Reference: Linear Algebra by S. Friedberg, A. Insel, and L. Spence

Linear Algebra and its Applications by G. Strang

4. Class Method

A. Lecture on new concepts

B. Practice problem sessions

C. Homework and quizzes

5. Evaluation

A. Evaluation Criteria

Activities(Homework, Quizzes, etc.)	Midterm exam	Final exam	Attendance and Attitude	Sum
40	60	80	20	200

B. Evaluation Method

(1) Homework

(2) Quizzes

(3) Exams

- Midterm Exam : Chapter 1-7

- Final Exam : Chapter 1-13

(4) Attendance

- **A 3-point deduction** for an absence and a **1-point deduction** for being tardy.

- No penalty will be given for official absence including illness.

6. Criteria for Awarding Credits

☞ **Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.**

- Attendance for at least three-quarters of the total class hours is required to earn credit.
 - Calculation of the basis score for awarding credit to absentees : In case of an examination missed due to official leave or medical leave, the score will be calculated by multiplying (the score corresponding to the total percentile score earned by the student on other examinations in the same subject) by (100% for official leave, or 80% for medical leave).
 - If a basis score cannot be calculated, the grading method : A make-up examination will be conducted.
- ※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 1 2.23.~3.1.	1.1 What is a matrix? 1.2 Matrix addition and scalar multiplication 1.3 Matrix multiplication 1.4 Matrix algebra 1.5 Matrix inverses 1.6 Powers of a matrix 1.7 The transpose and symmetric matrices 1.8 Vectors in $\mathbb{R}^n$ 1.9 Developing geometric insight 1.10 Lines 1.11 Planes in $\mathbb{R}^3$ 1.12 Lines and hyperplanes in $\mathbb{R}^n$ 2.1 Systems of linear equations 2.2 Row operations 2.3 Gaussian elimination 2.4 Homogeneous systems and null space	Self-Study on CH1 (Much of this has already been covered in Math2)
Core Competencies	◆ Know the essential terminologies and ways of solving linear equation	
Week 2 3.2.~3.8.	3.1 Matrix inverse using row operations 3.2 Determinants 3.3 Results on determinants	3.2. Substitute Holiday
Core Competencies	◆ Know the essential terminologies and how to calculate determinants	
Week 3 3.9.~3.15.	3.4 Matrix inverse using cofactors 3.5 Leontief input-output analysis * LU-decomposition	

	4.1 The rank of a matrix 4.2 Rank and systems of linear equations 4.3 Range	
Core Competencies	♦ Know the essential terminologies and how to find the rank of a matrix	
Week 4 3.16.~3.22.	5.1 Vector spaces 5.2 Subspaces 5.3 Linear span 6.1 Linear independence 6.2 Bases	
Core Competencies	♦ Know the essential terminologies	
Week 5 3.23.~3.29.	6.3 Coordinates 6.4 Dimension 6.5 Basis and dimension in $\mathbb{R}^n$ 7.1 Linear transformations	
Core Competencies	♦ Know the essential terminologies and generalize definitions	
Week 6 3.30.~4.5..	7.2 Range and null space 7.3 Coordinate change 7.4 Change of basis and similarity	
Core Competencies	♦ Know the essential terminologies and generalize definitions	
Week 7 4.6.~4.12.	Review	
Core Competencies	♦ Fully understand Chapter 1-7	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)	4.13. Regular Classes
Week 9 4.20.~4.26.	8.1 Eigenvalues and eigenvectors 8.2 Diagonalisation of a square matrix 8.3 When is diagonalisation possible?	
Core Competencies	♦ Know the essential terminologies and how to calculate eigenvalues	
Week 10 4.27.~5.3.	9.1 Powers of matrices 9.2 Systems of difference equations	

	9.3 Linear systems of differential equations 10.1 Inner products 10.2 Orthogonality	
Core Competencies	♦ Know the essential terminologies and how to solve equations	
Week 11 5.4.~5.10.	10.3 Orthogonal matrices 10.4 Gram-Schmidt orthonormalisation process * QR-decomposition 11.1 Orthogonal diagonalisation of symmetric matrices 11.2 Quadratic forms	5.5. Children's Day
Core Competencies	♦ Know the essential terminologies and how to orthogonalize matrices	
Week 12 5.11.~5.17.	12.1 The direct sum of two subspaces 12.2 Orthogonal complements 12.3 Projections 12.4 Characterising projections and orthogonal projections	
Core Competencies	♦ Learn the essential terminologies related to projection	
Week 13 5.18.~5.24.	12.5 Orthogonal projection onto the range of a matrix 12.6 Minimising the distance to a subspace 12.7 Fitting functions to data: least squares approximation 13.1 Complex numbers 13.2 Complex vector spaces	
Core Competencies	♦ Learn how to apply the terminologies to real world problems	
Week 14 5.25.~5.31..	13.3 Complex matrices 13.4 Complex inner product spaces 13.5 Hermitian conjugates	5.25. Substitute Holiday
Core Competencies	♦ Expand the notions from real numbers to complex numbers	
Week 15 6.1.~6.7.	13.6 Unitary diagonalisation and normal matrices 13.7 Spectral decomposition * Singular value decomposition	6.3. Local Election Day
Core Competencies	♦ Find the difference between real matrices and complex matrices	
Week 16 6.8.~6.14.	Make-up class week	

Week 17 6.15.~6.19.	Final Exams (6.18.~6.19.)	
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※ The above plans may change according to the academic schedule.